Toy-car data processing and interpretation

STUDENT WORKSHEET

Name: _____ Class: _____ Date: _____

Part A — position–time in Desmos

1. Copy your averaged (t, x) pairs into a Desmos table. Check both SI units and axis orientation.

2. Graph the raw points and add a suitable trend line/curve. Save a screenshot of both.

3. Identify and estimate an early slope and a late slope. Show how you obtained each, with units.

4. Explain the meaning of the slopes in terms of the car's motion. Compare to your manual results.

Part B — creating velocity–time data

5. Use your measured means to complete the table. For each consecutive pair, $\Delta x = 0.05$ m; Δt is the difference between their mean times. Calculate $v_{interval} = \Delta x/\Delta t$ and midpoint time.

Interval x (m)	Δx (m)	Δt (s)	Midpoint t (s)	Average v (m/s)
0.00–0.05	0.05			
0.05–0.10	0.05			
0.10–0.15	0.05			
0.15–0.20	0.05			

6. Plot average interval velocity (vertical) versus midpoint time (horizontal) in Desmos. Explain why this is an approximation.

7. Does your v–t graph suggest a roughly constant acceleration? Support your conclusion with graphical evidence.

8. How does velocity change (a) per centimetre of travel and (b) per second? Explain why the two questions are different.

9. Identify one limitation you noticed during the experiment and propose an improvement with reasoning.

Homework — Criterion B: review a plan

Read the separate "Criterion B practice: toy-car plan". For each of A1–A4, identify strengths, identify missing/unclear details and suggest specific improvements. Propose a provisional achievement level based on the rubric supplied by your teacher.